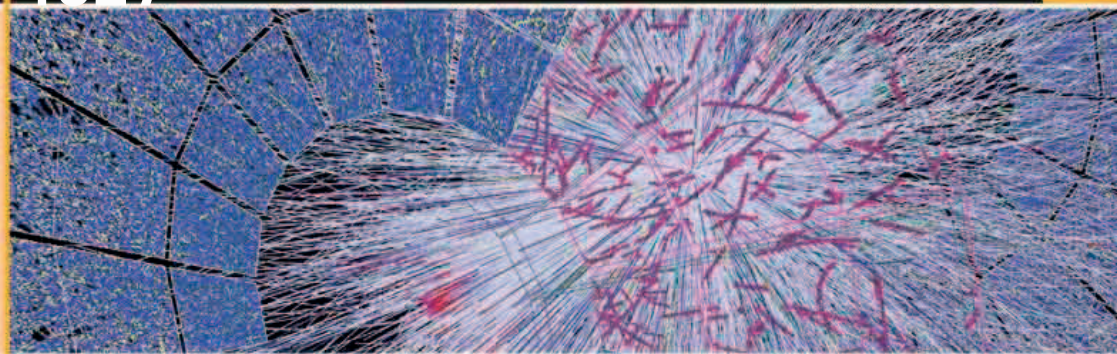
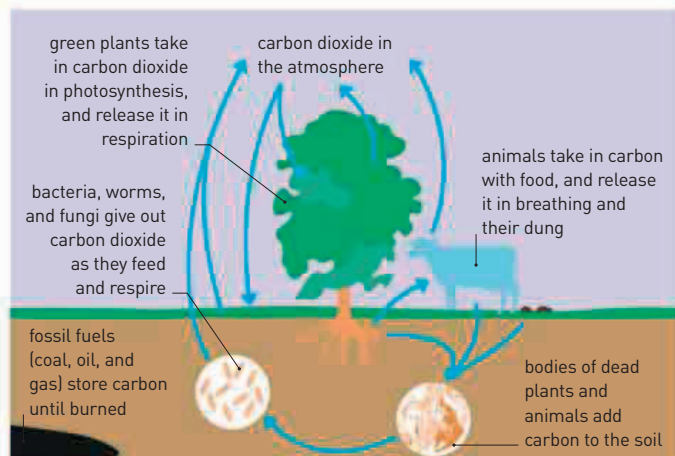


“ I WAS... ABLE TO **TRANSMIT THE LIVING IMAGE**, AND IT WAS THE **FIRST TIME** IT HAD BEEN DONE. BUT HOW TO CONVINCE THE... SCEPTICAL SCIENTIFIC WORLD? ”

John Logie Baird, Scottish engineer, in *The Times*, January 28, 1926



A computer-generated simulated image of a particle collision shows matter thought to have been produced microseconds after the Big Bang.



THE CARBON CYCLE

At the heart of Vladimir Vernadsky's book *The Biosphere* are his ideas about nature's constant recycling of matter. Elemental atoms react and recombine in different ways. Carbon atoms in complex organic materials of living things—animals, plants, and soil bacteria—are respired into the atmosphere as carbon dioxide, before reacting to make plant sugars in photosynthesis.

appearance in its modern form.

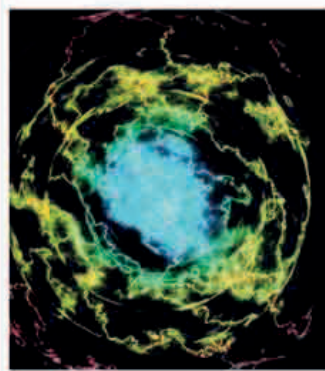
Chemists had been making this chemical polymer since the 1800s, but American chemist **Waldo Semon** (1898–1999) found a way of making it malleable and less brittle. It would become one of the most widely used plastics.

Russian geochemist **Vladimir Vernadsky** (1863–1945) had been working on theories that unified geology and biology, Earth and life. He published his thoughts in a book called *The*

Biosphere. In it he described some of the main ideas behind a concept today referred to as the **ecosystem: a system where living things interact with nonliving matter**. Vernadsky recognized that all life on Earth **relies upon solar energy**, and that particles of matter **undergo a process of recycling**.

THE YEAR STARTED WITH A MILESTONE in communication technology. On January 7, a collaboration between the American Telephone and Telegraph Company and Britain's General Post Office **opened the first transatlantic telephone service**. On its first day 31 calls were made between London and New York.

A further development came in the field of quantum physics. Erwin Schrodinger had laid the foundations of quantum mechanics with his description of the wavelike characteristics of particles. Now, German physicist **Werner Heisenberg** (1901–76) reasoned that the wave function of a particle could not be localized to a specific point in space and have a definable wavelength. Heisenberg developed this as his **Uncertainty Principle**. Its consequences are extraordinary:



the more accurately a particle's position is measured, the less accurately it is possible to determine its movement, and *vice versa*. Later, the Copenhagen Interpretation stated that it is impossible to experimentally measure wavelike and particle-like properties at the same time.

While Baird worked in London on his television, the **American Bell Telephone Company** was also developing the technology. In April, Bell had a breakthrough when the company sent the first **long-distance TV transmission** using the semimechanical television from Washington to New York. Five months later, American inventor **Philo Farnsworth** (1906–71) introduced a way of scanning and transmitting electronically. Russian-American inventor **Vladimir Zworykin** (1888–1982) was working on similar technology at the same time, but it was Farnsworth who made it a reality. In April, Belgian astronomer **Georges Lemaître** (1894–1966) published a scientific paper containing a revolutionary theory: that the **Universe is expanding**.

Electron clouds

Modern quantum physics has revised the idea that atoms—such as this helium atom—have electrons in fixed orbits. A more realistic interpretation sees electrons existing as clouds of probability.

When Lemaître presented his conclusions to the British Association for the Advancement of Science in 1931, he elaborated upon his theory, suggesting the Universe had originated from a primeval atom. His “exploding cosmic egg” model anticipated the work of American astronomer Edwin Hubble (see 1929) and was the forerunner of the Big Bang theory.



WERNER HEISENBERG
(1901–76)

Heisenberg studied physics at the universities of Munich and Göttingen, where he met Niels Bohr in 1922. In 1925, he developed a mathematical way of understanding quantum physics called matrix mechanics. He derived his Uncertainty Principle before working on the German nuclear energy project in World War II.

